

✓ 1.1

1. $63_{10} \rightarrow \text{BIN}$

63	2
62	31
1	30
	15
	14
	1
	7
	6
	1
	2
	3
	2
	1
	2
	1
	0
	0
	1

Jawab: $63_{10} = 111111_2$

2. $63_{10} \rightarrow \text{HEX}$

63	16
48	3
15	0
	0
	3

Jawab: $63_{10} = 3F_{16}$

3. $170_{10} \rightarrow \text{BIN}$

170	2
170	85
0	84
	42
	1
	42
	21
	0
	20
	10
	1
	10
	0
	5
	4
	1
	2
	2
	2
	1
	0
	0
	1

Jawab: $170_{10} = 10101010_2$

4. $170_{10} \rightarrow \text{HEX}$

$$\begin{array}{r|l} 170 & 16 \\ -160 & 10 \\ \hline 10 & 0 \\ & 10 \end{array}$$

Answer: $170_{10} = AA_{16}$

✓ 1.2

1. $11101001_2 \rightarrow \text{DEC}$

$$\begin{array}{cccccccc} 7 & 6 & 5 & 4 & 3 & 2 & 1 & 0 \\ 1 & 1 & 1 & 0 & 1 & 0 & 0 & 1 \end{array} = 1 \cdot 2^7 + 1 \cdot 2^6 + 1 \cdot 2^5 + 0 \cdot 2^4 + 1 \cdot 2^3 + 0 \cdot 2^2 + 0 \cdot 2^1 + 1 \cdot 2^0$$
$$= 128 + 64 + 32 + 0 + 8 + 0 + 0 + 1 = 233$$

Answer: $11101001_2 = 233_{10}$

2. $11101001_2 \rightarrow \text{HEX}$

1110 1001

$$\begin{array}{cccc} 3 & 2 & 1 & 0 \\ 1 & 1 & 1 & 0 \end{array} = 1 \cdot 2^3 + 1 \cdot 2^2 + 1 \cdot 2^1 + 0 \cdot 2^0 = 8 + 4 + 2 + 0 = 14_{10} = E_{16}$$

$$\begin{array}{cccc} 3 & 2 & 1 & 0 \\ 1 & 0 & 0 & 1 \end{array} = 1 \cdot 2^3 + 0 \cdot 2^2 + 0 \cdot 2^1 + 1 \cdot 2^0 = 8 + 0 + 0 + 1 = 9_{10} = 9_{16}$$

Answer: $11101001_2 = E9_{16}$

✓ 1.3

$$1000111_2 + 11100_2 = x_2$$

$$\begin{array}{r} 111 \\ + 1000111 \\ \hline 0011100 \\ \hline 1100011 \end{array}$$

Answer: 1100011_2